

This chapter contains an overview of the topographical anatomy of the head and neck described in detail in [Chapters 27–42](#), and an account of the clinically relevant surface anatomy.

SKIN AND FASCIA

FACE AND SCALP

The superficial fascia of the scalp is firm, dense, fibroadipose, and closely adherent to the skin and to the underlying muscle, epicranium and the epicranial aponeurosis. Posteriorly, the fascia is continuous with the superficial fascia of the back of the neck, and laterally it is prolonged into the temporal region, where it is looser in texture. Three fascial layers (a subcutaneous fibroadipose tissue, a superficial musculo-aponeurotic system (SMAS) and the parotid–masseteric fascia) are recognized on the face superficial to the plane of the facial nerve and its branches. On the lateral side of the head, above the zygomatic arch, the temporoparietal fascia lies in the same plane as, but does not blend with, the superficial musculo-aponeurotic system. It is superficial to the temporal fascia and blends superiorly with the epicranial aponeurosis. The parotid gland is surrounded by a fibrous capsule derived from the deep cervical fascia.

NECK

The superficial cervical fascia is a zone of loose connective tissue between dermis and deep fascia, and is joined to both. It contains a variable amount of adipose tissue and platysma but is hardly demonstrable as a separate layer. The deep cervical fascia is conventionally subdivided into three sheets (superficial investing, middle and deep layers) that surround the muscles and viscera of the neck to varying degrees, and the carotid sheath, a condensation of deep fascia around the common and internal carotid arteries, internal jugular vein, vagus and ansa cervicalis. The fascial layers of the neck define a number of potential tissue ‘spaces’ above and below the hyoid bone ([Ch. 31](#)). In health, the tissues within these spaces are either closely applied to each other or are filled with relatively loose connective tissue. However, they offer potential routes by which unchecked infection may spread within the head and neck, and between the tissue spaces of the face and the mediastinum. They also offer convenient planes for dissection during surgery.

BONES AND JOINTS

SKULL AND MANDIBLE

The skull is composed of 28 separate bones, most of them paired ([Ch. 27](#)). It can be divided into the cranium, consisting of the calvaria (brain box) and basicranium, which together surround and protect the brain; a delicate facial skeleton composed mainly of thin-walled bones, some of which contain air-filled cavities that are known collectively as the paranasal sinuses; and the mandible. The cranial cavity contains the brain and the intracranial portions of the cranial nerves; the blood vessels that supply and drain the brain and the haemopoietic marrow of the overlying bones; the meninges (dura, arachnoid and pia mater); and the cerebrospinal fluid in the subarachnoid space. The cavity is incompletely divided by dural partitions, notably the falx cerebri, lying between the cerebral hemispheres, and the tentorium cerebelli, lying between the cerebellum and occipital lobes. Almost all of the venous blood from the brain and cranial bones drains via sinuses lying between the endosteal and meningeal layers of the dura mater into the internal jugular vein. Internally, the cranial base is divided into anterior, middle and posterior cranial fossae, which contain the frontal and temporal

lobes of the cerebral hemispheres and the cerebellum, respectively. Foramina in the bones of the skull base and facial skeleton transmit neurovascular bundles that may be compromised at these sites by pathology or trauma.

The bony orbits contain the eyeballs, oculogyric muscles and lacrimal glands with their associated neurovascular supplies ([Chs 41 and 42](#)). The temporal bones contain the inner, middle and external ears ([Chs 37 and 38](#)). The maxillae are the largest of the pneumatized bones of the midface; they contain the maxillary air sinuses and bear the upper teeth. The mandible bears the lower teeth and articulates with the temporal bones at the temporomandibular joints. The skull articulates with the first cervical vertebra (atlas); movements of the skull on the cervical vertebrae occur at the atlanto-occipital joints.

The skull provides attachments for many muscles, including all the craniofacial muscles, the oculogyric muscles, the muscles that act on the temporomandibular joint, the superior constrictor of the pharynx, the muscles of the soft palate, all but one of the extrinsic muscles of the tongue, the muscles of the suboccipital region, and the cranial attachments of trapezius and sternocleidomastoid.

CERVICAL VERTEBRAE

There are seven cervical vertebrae. They are the smallest of the movable vertebrae and are characterized by a disproportionately large vertebral canal. All but the seventh are also characterized by a foramen in each transverse process, the foramen transversarium (these foramina may be absent or sometimes duplicated in the seventh cervical vertebra). The first (atlas), second (axis) and seventh (vertebra prominens) cervical vertebrae are atypical.

HYOID BONE AND LARYNGEAL CARTILAGES

The hyoid bone lies in the midline at the front of the neck at the level of the third cervical vertebra. It is suspended from the styloid processes by the stylohyoid ligaments and gives attachment to the suprahyoid and infrahyoid groups of muscles. The skeletal framework of the larynx is formed by a series of cartilages interconnected by ligaments and fibrous membranes, and moved by a number of muscles ([Ch. 35](#)). The laryngeal cartilages are the single cricoid, thyroid and epiglottic cartilages, and the paired arytenoid, cuneiform, corniculate and tritiate cartilages.

MUSCLES

The striated muscles of the head and neck produce the movements of the facial soft tissues that animate so many aspects of communication; the movements at the temporomandibular joint that occur during mastication and speech ([Ch. 32](#)); the conjugate movements of the eyeballs; and the coordinated movements that occur during activities such as swallowing, speaking and turning the head in response to visual and/or auditory stimuli. The ‘extrinsic’ muscles that run between the axial skeleton and upper limb act on the scapula and humerus ([Ch. 48](#)).

The superior tarsal muscle, sphincter and dilator pupillae and the ciliary muscle are composed of smooth muscle ([Chs 41 and 42](#)).

VASCULAR SUPPLY AND LYMPHATIC DRAINAGE

ARTERIES

The main arterial supply of the head and neck is derived from branches of the carotid and subclavian arteries ([Figs 26.1–26.2](#)). Branches of the

Fig. 26.1 An overview of the superficial arteries of the head and neck.

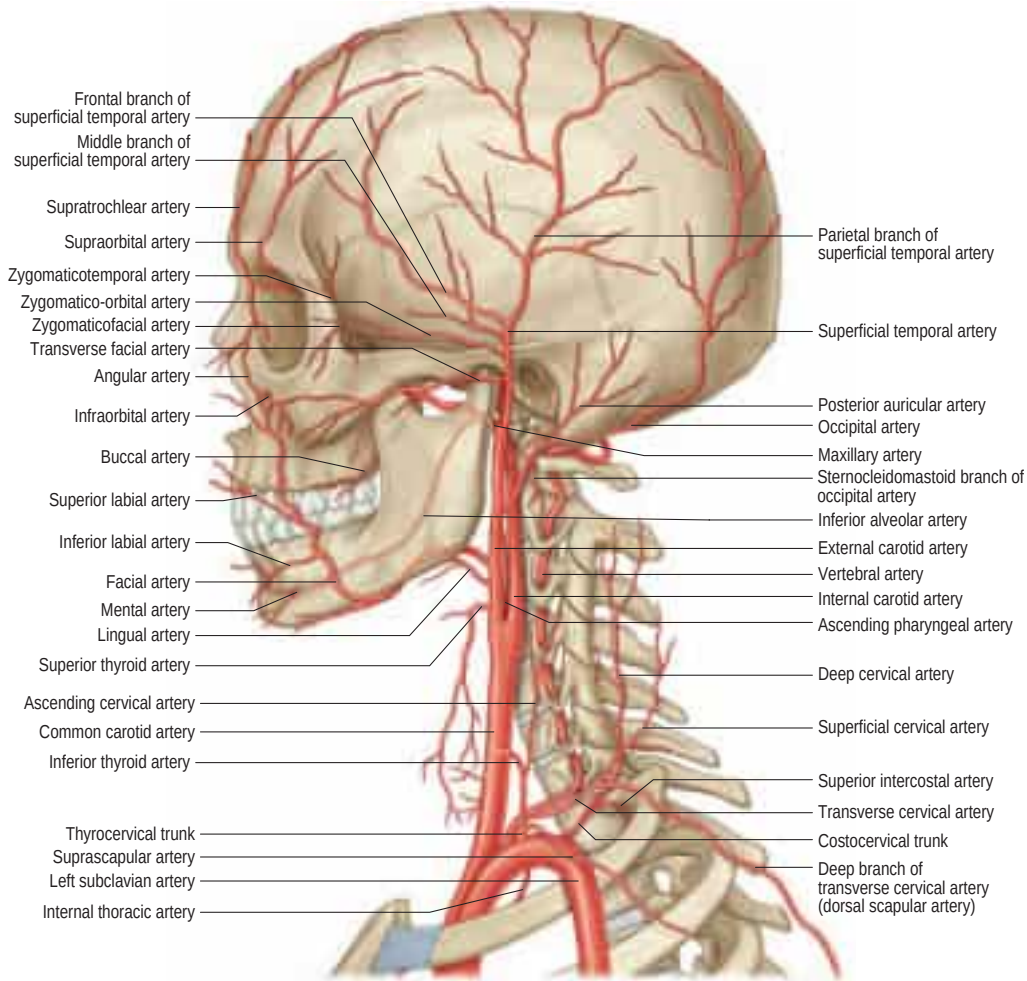
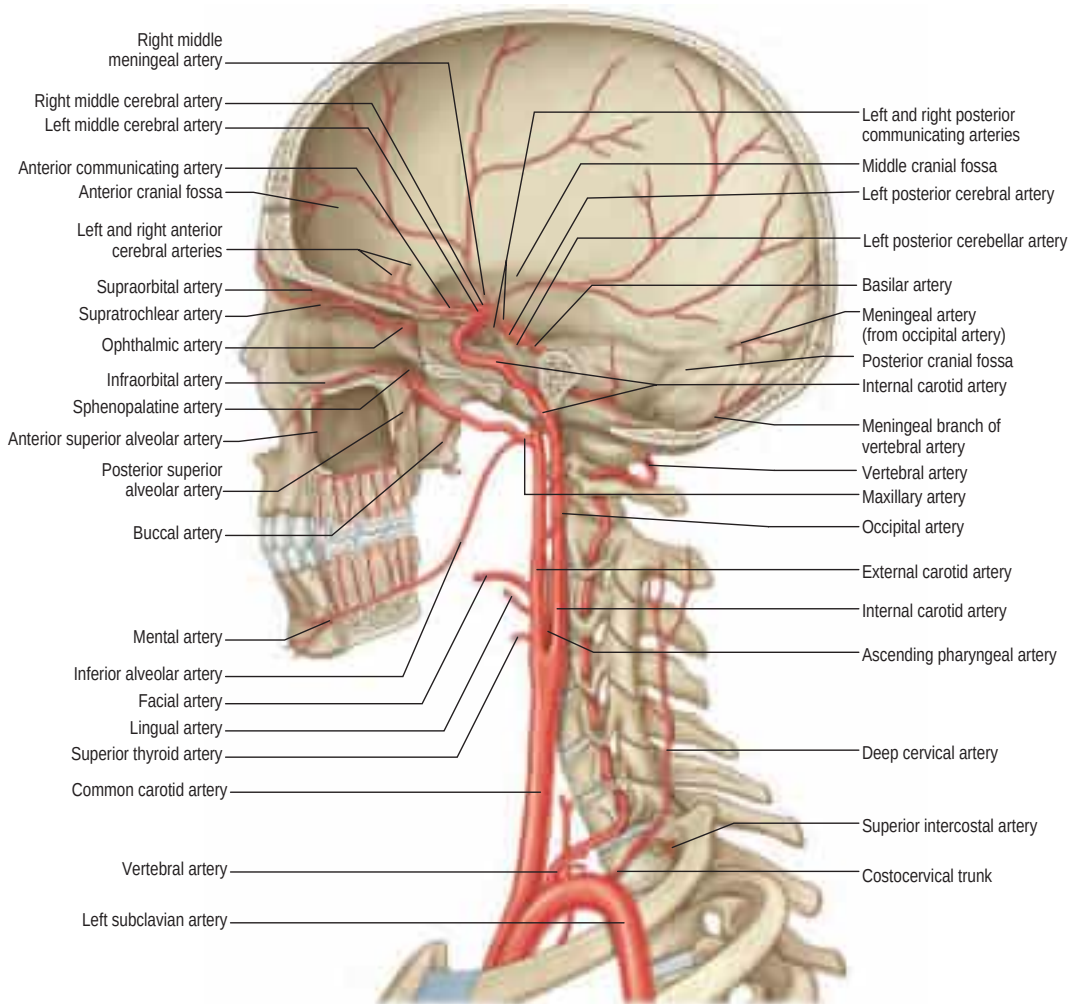


Fig. 26.2 An overview of the deep arteries of the head and neck.



internal carotid and vertebral arteries anastomose in the circle of Willis within the interpeduncular cistern on the ventral aspect of the brain (Ch. 19).

Carotid system

The cervical portion of the common carotid artery is similar on both sides. Each lies within the carotid sheath of deep cervical fascia, together with the internal jugular vein and vagus nerve. In the lower part of the neck, the arteries are separated by a narrow gap that contains the trachea, and higher up they are separated by the thyroid gland, larynx and pharynx. At the level of the upper border of the thyroid cartilage (C4), the common carotid artery bifurcates into external and internal carotid arteries (see below). The external carotid artery passes upwards on either side of the neck, inclined at first slightly forwards and then backwards and a little laterally. It usually gives off the ascending pharyngeal, superior thyroid, lingual, facial, occipital and posterior auricular arteries, and then enters the parotid salivary gland where it divides into its terminal branches, the superficial temporal and maxillary arteries. The branches of the external carotid artery supply the face, scalp, tongue, upper and lower teeth and gingivae, palatine tonsil, paranasal sinuses and nasopharyngeal tube, external and middle ears, pharynx, larynx and superior pole of the thyroid gland. They also anastomose with branches of the internal carotid arteries on the scalp, forehead and face, in the orbit, nasopharynx and nasal cavity, and with branches of the subclavian artery in the pharynx, larynx and thyroid glands.

The internal carotid artery supplies most of the ipsilateral cerebral hemisphere, eye and accessory organs, the forehead and, in part, the external nose, nasal cavity and paranasal sinuses. It passes up the neck anterior to the transverse processes of the upper three cervical vertebrae and enters the cranial cavity via the carotid canal in the petrous part of the temporal bone. The artery has no branches in the neck and so is easily distinguishable from the external carotid artery, should the latter require ligation, e.g. to control haemorrhage from a penetrating injury to the neck.

Subclavian artery

The subclavian arteries give off several branches that supply structures in the head and neck. The vertebral arteries supply the upper spinal cord, brainstem, cerebellum and occipital lobe of the cerebrum. They pass through the foramina transversaria of the first six cervical vertebrae, enter the cranial cavity through the foramen magnum and unite at the lower border of the pons to form the basilar artery (hence this system is often called the vertebrobasilar system). Branches from the thyro-cervical trunk supply the inferior poles of the thyroid gland and the parathyroid glands, the larynx and the pharynx, and branches from the costocervical trunks supply deep cervical muscles.

VEINS

The veins of the neck lie superficial or deep to the deep investing fascia. Superficial veins ultimately drain into either the external, anterior or posterior external jugular veins; they drain a much smaller volume of tissue than the deep veins. Deep veins tend to drain into either the internal jugular vein or the subclavian vein. The internal jugular vein drains blood from the skull, brain, superficial face and much of the neck. It descends in the neck within the carotid sheath and unites with the subclavian vein behind the sternal end of the clavicle to form the brachiocephalic vein (Figs 26.3–26.4). At its junction with the internal jugular vein, the left subclavian vein usually receives the thoracic duct, and the right subclavian vein receives the right lymphatic duct.

LYMPHATIC DRAINAGE

Lymph nodes in the head and neck are arranged in two horizontal rings and two vertical chains on either side of the neck (Fig. 26.5). The outer, superficial, ring consists of the occipital, preauricular (parotid), submandibular and submental nodes, and the inner, deep, ring is formed by clumps of mucosa-associated lymphoid tissue (MALT) located

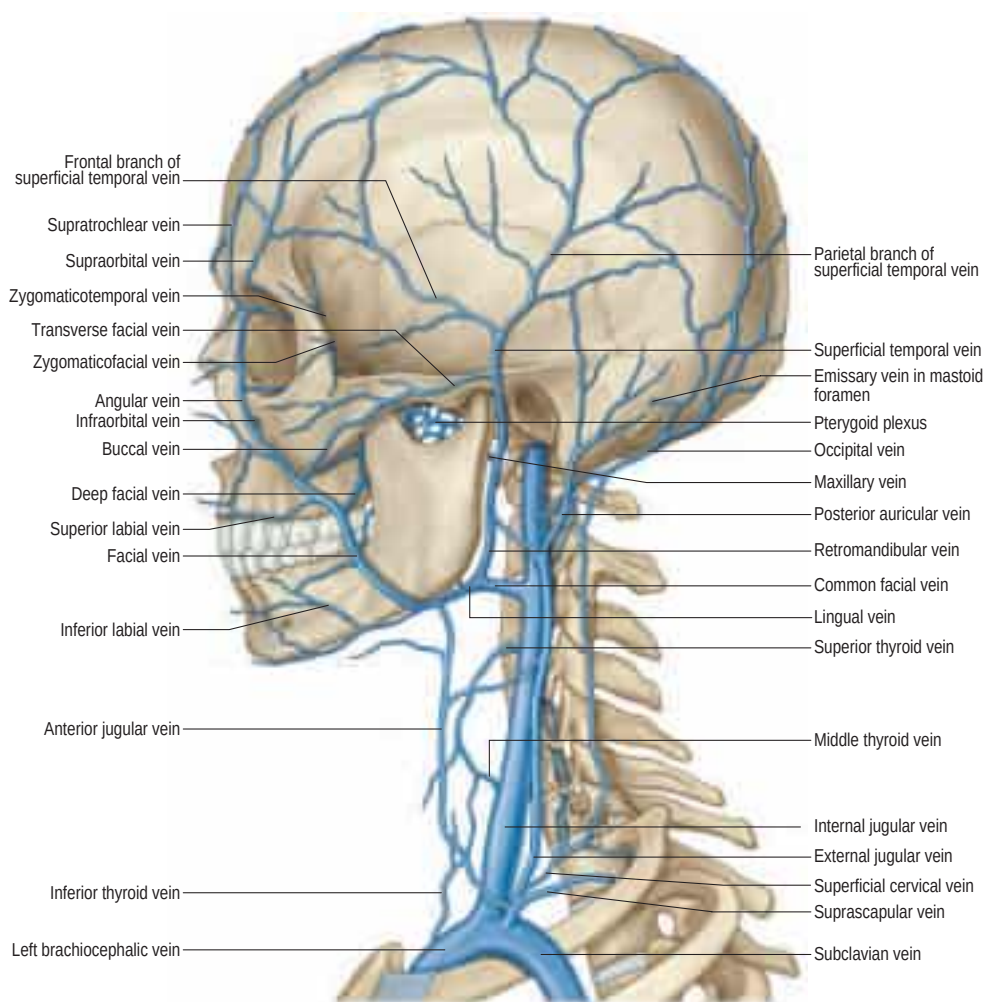


Fig. 26.3 An overview of the superficial veins of the head and neck.

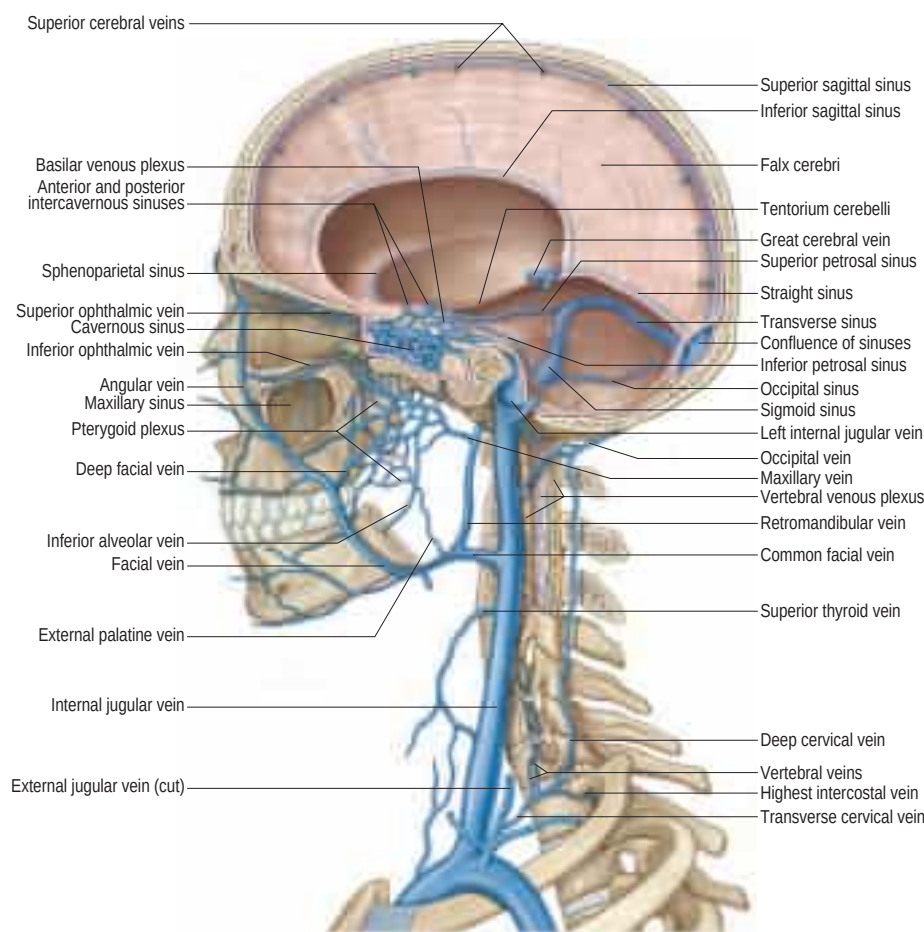


Fig. 26.4 An overview of the deep veins of the head and neck.

primarily in the nasopharynx and oropharynx (Waldeyer's ring). The vertical chain consists of superior and inferior groups of nodes related to the carotid sheath. All lymph vessels of the head and neck drain into the deep cervical nodes, either directly from the tissues or indirectly via nodes in outlying groups. Lymph is returned to the systemic venous circulation via either the right lymphatic duct or the thoracic duct.

INNERVATION

CRANIAL NERVES

There are 12 pairs of cranial nerves. They are individually named and numbered (using Roman numerals) in a rostrocaudal sequence (see Table 16.1). Some are functionally mixed, others are either purely motor or purely sensory, and some also carry pre- or postganglionic parasympathetic fibres that are secretomotor to the salivary and lacrimal glands or motor to the smooth muscle within the eyeball and orbit. Branches of the oculomotor, trochlear, trigeminal, abducens, facial, glossopharyngeal, vagus, accessory and hypoglossal nerves supply muscle groups within the eyeball, face, neck, pharynx, larynx and tongue. Branches of the trigeminal, glossopharyngeal and vagus nerves transmit general sensory information from the skin of the face and part of the scalp; the epithelium lining the oral and nasal cavities, the paranasal sinuses, middle ear, pharynx and larynx, and the dorsal surface of the tongue and the cornea; the intracranial meninges; and the periosteum and bones of the skull. Branches of the trigeminal nerve innervate the temporomandibular joint. The olfactory, optic, trigeminal, facial, vestibulocochlear and vagus nerves contain axons that transmit the special sensations of olfaction, vision, hearing, balance and taste. The olfactory nerve is the only sensory cranial nerve that projects directly to the cerebral cortex rather than indirectly via the thalamus (Ch. 25). The optic nerve terminates in the thalamus (Ch. 23). The other ten pairs of cranial nerves are attached to the brainstem or, in the case of the accessory nerve, to the upper cervical spinal cord; their component fibres arise from or terminate in named cranial nerve nuclei. The cranial nerves pass through named foramina in the skull, often with named vessels.

With one exception, all of the cranial nerves are confined to the head and neck. The exception is the vagus, which travels through the neck

and thorax, and enters the abdominal cavity by passing through the diaphragm with the oesophagus.

Reflexes

A number of reflexes involving structures in the head and neck are mediated by sensory and motor branches of certain of the cranial nerves, coordinated via appropriate nuclei in the brainstem. They include swallowing, gagging, retching and vomiting, sneezing and coughing; lacrimation; jaw jerk; visual reflexes (pupillary light reflex and accommodation); and the corneal 'blink' reflex and the stapedial reflex. Reflexes that involve energetic exhalation, e.g. sneezing and coughing, also involve the recruitment of cervical and thoracic spinal neurones to mediate the coordinated contraction of intercostal and abdominal wall muscles that this activity requires.

SPINAL NERVES

There are eight pairs of cervical spinal nerves (Fig. 26.6). Cutaneous branches of the dorsal rami of the second, third, fourth and fifth cervical nerves innervate the scalp and the skin over the back of the neck, and motor branches of all of the cervical dorsal rami supply cervical post-vertebral muscles. All of the cervical ventral rami supply anterior and lateral groups of prevertebral muscles. The upper four cervical ventral rami form the cervical plexus, whose branches collectively innervate the infrahyoid strap muscles and the diaphragm, and the skin covering the lateral and anterior parts of the neck, and the angle of the mandible. The lower four cervical ventral rami, together with most of the first thoracic ventral ramus, form the brachial plexus.

PARASYMPATHETIC NERVES

In the head and neck, the parasympathetic system innervates the salivary and lacrimal glands, the mucous glands of the oral and nasal cavities and paranasal sinuses, and the sphincter pupillae and ciliary muscles in the eyeball. Many thousands of preganglionic parasympathetic axons travel in the vagus from cell bodies in the dorsal motor nucleus of the vagus in the medulla. Although they travel through the



There are four pairs of parasympathetic ganglia in the head, named ciliary, pterygopalatine, otic and submandibular (**Fig. 26.7**).

Preganglionic axons originate in the Edinger–Westphal preganglionic nucleus of the midbrain (Ch. 21). They travel via a branch of the oculomotor nerve (nerve to the inferior oblique) to the ciliary ganglion, where they synapse. Postganglionic fibres travel in the short ciliary nerves, which pierce the scleral coat of the eyeball and run forwards in the perichoroidal space to enter the ciliary muscle and sphincter pupillae. Their activation mediates accommodation of the eye to near objects and pupillary constriction.

Preganglionic axons originate in the superior salivatory nucleus. They emerge from the brainstem in the nervus intermedius and leave the main facial nerve trunk in the middle ear to join the chorda tympani, which subsequently joins the lingual nerve. In this way, they reach the submandibular ganglion, where they synapse. Postganglionic fibres innervate the submandibular, sublingual and lingual salivary glands; some axons presumably re-enter the lingual nerve to access the lingual glands, while others pass directly along blood vessels to enter the submandibular and sublingual glands.

Pterygopalatine ganglion

Otic ganglion

Preganglionic axons originate in the inferior salivatory nucleus and travel in the glossopharyngeal nerve and its tympanic branch. They traverse the tympanic plexus and lesser petrosal nerve, and pass through the foramen ovale to reach the otic ganglion, where they synapse. Postganglionic fibres pass by communicating branches to the

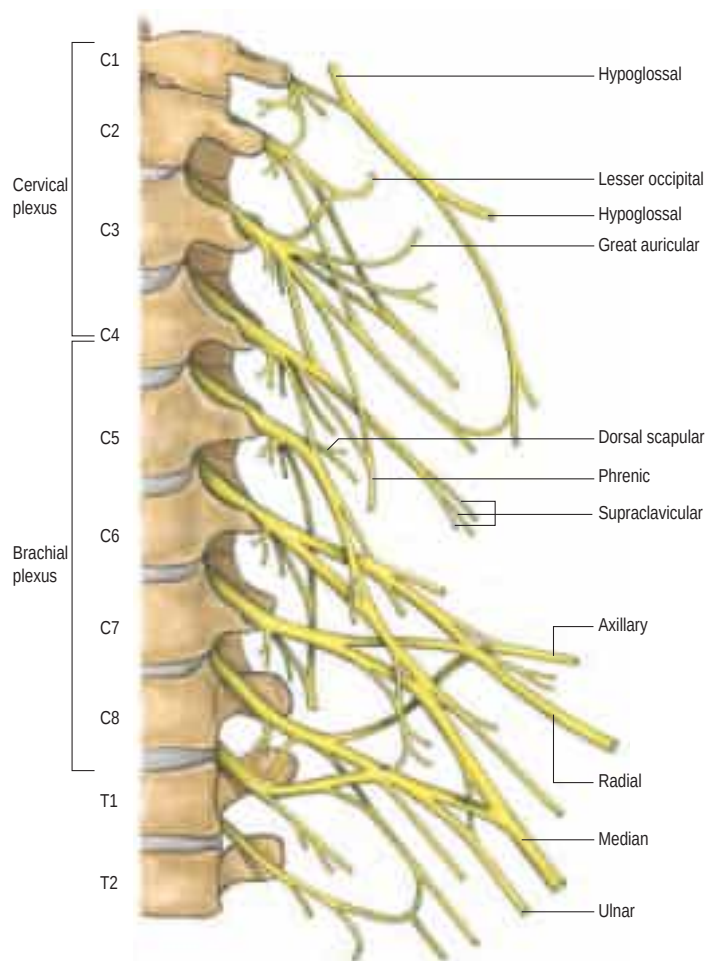


Fig. 26.6 An overview of the nerves forming the cervical and brachial plexuses. (With permission from Waschke J, Paulsen F (eds), Sobotta Atlas of Human Anatomy, 15th ed, Elsevier, Urban & Fischer. Copyright 2013.)

auriculotemporal nerve, which conveys them to the parotid gland. Stimulation of the lesser petrosal nerve produces vasodilator and secretomotor effects.

SYMPATHETIC NERVES

In the neck, the sympathetic trunk lies behind and medial to the carotid sheath and anterior to the transverse processes of the cervical vertebrae and the prevertebral muscles. There are usually three cervical ganglia on each side – superior, middle and inferior (cervicothoracic); they may be connected by a solid trunk or by two or three fibrous strands. The ganglia receive preganglionic fibres from neurones whose cell bodies lie in the intermediolateral column of the upper thoracic spinal cord; there is no preganglionic output from the cervical spinal cord. Postganglionic fibres reach their target tissues in the head and neck via the cervical spinal nerves and perivascular nerve plexuses distributed along the carotid and vertebral arteries (see Fig. 26.7).

SURFACE ANATOMY

SURFACE ANATOMY OF THE HEAD

Skeletal landmarks

The palpable bony landmarks of the head can be examined from the back, sides and front (Figs 26.8–26.9). The pericraniocervical line, which demarcates the head from the neck, runs from the midpoint of the chin anteriorly to the external occipital protuberance posteriorly.

Calvaria

Most of the superficial aspect of the calvaria is covered by skin, subcutaneous tissue and thin muscles; many bony prominences and surfaces are therefore palpable through the hair and are visible in bald people.

Suture lines can be palpated as irregular ridges. The anterior fontanelle, at the junction of the coronal and sagittal sutures, may be palpated up to 18 months after birth. The mastoid process is palpable medial to the lobule of the pinna. Deep palpation immediately anteroinferior to the mastoid apex reveals the relatively indistinct resistance offered by the styloid process and associated soft tissues, and posteroinferior palpation reveals the firm mass of the transverse process of atlas (C1). The external occipital protuberance (inion) is a palpable bony protuberance located on the posterior midline of the occipital bone. The confluence of the dural venous sinuses sits mainly to the upper right side of the inion (Sheng et al 2012). The superior nuchal lines are palpable passing laterally from the inion. The asterion can be palpated as a slight depression 1–2 cm behind the pinna at a level approximately at the junction of the upper third and lower two-thirds of the pinna. It marks the junction of the occipitomastoid, parietomastoid and occipitoparietal sutures, and is commonly employed to define the location of transverse–sigmoid sinus junction (Day and Tschabitscher 1998), although it does not always do so (Sheng et al 2012).

Mandible

The mental tubercles are palpable on the anterior chin, and the mandibular body, alveolar processes and teeth are palpable through the cheek. The lower border of the mandible can be traced to the angle at vertebral level C2. The mental foramen, which transmits the mental nerve and vessels, gradually migrates posteriorly after birth and through childhood (Balcioglu et al 2011). In adults, it lies an average of 2.5 cm from the midline face (Udhaya et al 2013), between the root apices of mandibular teeth 4 and 5 (range: teeth 3–6) (Fishel et al 1976, von Arx et al 2013, Udhaya et al 2013). Most foramina usually sit either level with or inferior to the tooth root apex (Fishel et al 1976), and accessory mental foramina, if present, are mainly situated posterior or inferior to the main foramen (Katakami et al 2008). The mandibular foramen sits on the medial surface of the ramus an average of 1.9 cm posterior to the third molar (Nicholson 1985); a 21 mm needle is therefore sufficient to reach the foramen for dental anaesthesia (Menke and Gowgiel 1979). The foramina almost always (Mbajjorgu and Ekanem 2000, Nicholson 1985) sit level with or below the occlusal plane, the remainder sitting above.

The mandibular ramus is largely covered by masseter and the parotid gland. Its posterior border is palpable through the parotid up to the neck of the mandibular condylar process, which lies anterior to the lower tragus/intertragal notch of the pinna. Movement of the condylar head at the temporomandibular joint is palpable anterior to the tragus during mouth opening/closing. The coronoid process is palpable anterior to the condylar process through the buccal mucosa.

Face

The lateral aspect of the face, from above downwards, consists of the temporal region, cheek and lower jaw (Ch. 30). The temporal region lies in front of the external ear and above the zygomatic arch. It is demarcated superiorly and posteriorly by the palpable superior temporal line, anteriorly by the frontal process of the zygoma, and inferiorly by the zygomatic arch. The palpable zygomatic arch is formed by the zygomatic process of the temporal bone posteriorly and the temporal process of the zygomatic bone anteriorly, both of which articulate via the palpable zygomaticotemporal suture. The variable prominence of the ‘cheekbone’ (zygoma) is largely attributable to the shape of the body of the underlying zygomatic bone. The sharp posterior margin of the frontal process of the zygomatic bone can be palpated superiorly to its articulation with the zygomatic process of the frontal bone at the palpable zygomaticofrontal suture. From here, the superior temporal line can be palpated passing posteriorly in the line of a gentle curve. The lower temporal line terminates by curving downwards and forwards to end just above the root of the mastoid process as the supramastoid crest on the squamous part of the temporal bone. The suprameatal triangle overlies the lateral wall of the mastoid (tympanic) antrum and is demarcated superiorly by the palpable supramastoid crest (a palpable ridge of bone posterior to the upper pinna), anteroinferiorly by the posterosuperior margin of the external acoustic meatus, and posteriorly by a vertical line through the posterior border of the external acoustic meatus. The lateral/Sylvian fissure of the brain aligns with the anterior part of squamosal suture in a zone 2.5–4.0 cm anterior to the external acoustic meatus (Rahmah et al 2011). The pterion represents the junction of the frontal, sphenoid, parietal and temporal sutures within the temporal fossa. It is situated within a 1 cm diameter circle centred approximately 2.6 cm posterior and 1.3 cm superior to the postero-lateral margin of the zygomaticofrontal suture (Ma et al 2012). Alternatively, the pterion is located approximately 4 cm above the midpoint of the zygomatic arch (Ukoha et al 2013). It usually marks the position

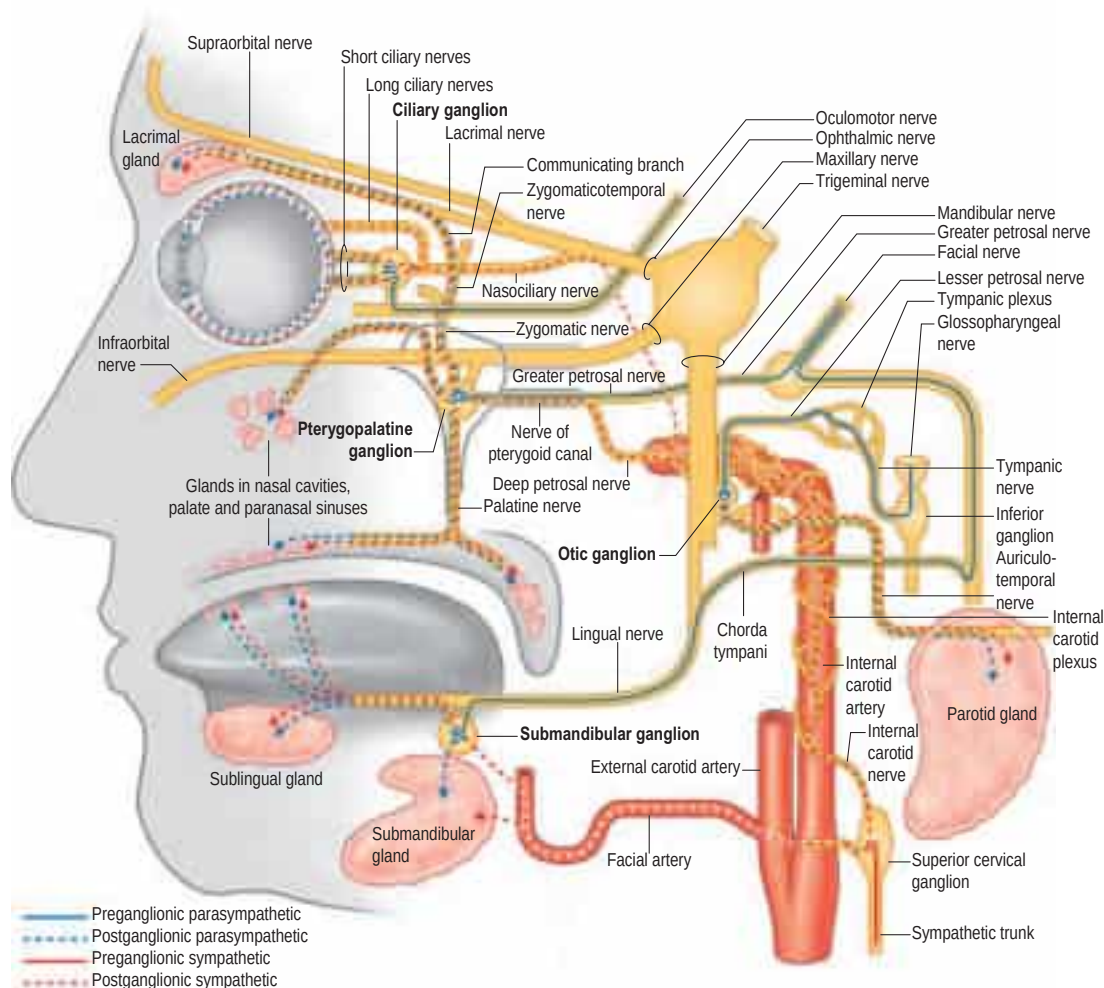


Fig. 26.7 An overview of the visceral efferent pathways in the head. (Adapted from Drake RL, Vogl AW, Mitchell A (eds), *Gray's Anatomy for Students*, 2nd ed, Elsevier, Churchill Livingstone. Copyright 2010.)

of the anterior (frontal) branch of the middle meningeal artery, with the remainder of arteries sitting immediately posterior ([Ma et al 2012](#)) ([Ch. 28](#)). Knowledge of middle meningeal artery surface anatomy is important for accurate positioning of burr-holes to evacuate extradural haematomas. The posterior (parietal) arterial branch runs backwards and parallel with the upper border of the zygomatic arch and is usually exposed vertically above the mastoid process on a level horizontal with the upper margin of the orbit. The asterion is a craniometric point at the site of the posterolateral, or mastoid, fontanelle. It is found at the junction of the lambdoid, occipitomastoid and parietomastoid sutures. It has been used as a landmark in lateral approaches to the posterior fossa as it defines the superior limit of bone removal for the craniectomy. However, the asterion almost always sits over the junction between the transverse and sigmoid sinuses, or sometimes inferior to the sinus junction ([Sheng et al 2012](#)).

The forehead extends from the hair margin of the scalp to the eyebrows. The rounded frontal tuberosity may be felt approximately 3 cm above the midpoint of each supraorbital margin. The superciliary arch is usually palpable above the orbit and is better marked in the male than the female. The glabella, a small horizontal ridge, is easily palpable between the superciliary arches. Below the glabella, the nasal bones meet the frontal bone in a small depression, the nasion, at the root of the nose. With a little finger inserted into the nostril, the bony margins of the anterior nasal aperture can be felt; they are formed by the inferior border of the nasal bone, the sharp margins of the nasal notch, and the coapted nasal spines of the maxillae ([Ch. 33](#)).

The orbital opening is relatively quadrangular and its margins palpable ([Ch. 41](#)). The supraorbital margin is formed entirely by the frontal bone. The supraorbital foramen/notch can be palpated along the superomedial orbital margin and sits an average of 2.7 cm (range 1.8–4.3 cm) from the midline face ([Chrcanovic et al 2011](#)). Foramen position can vary with gender, race, side and climatic origin ([Chrcanovic et al 2011](#), [Tomaszewska et al 2012](#)). This notch/foramen transmits the supraorbital nerve and vessels, and pressure exerted here with the fingernail can be painful. An additional notch may be found more towards the bridge of the nose; it transmits the supratrochlear neurovascular bundle.

The lateral margin of the orbit consists of the frontal process of the zygomatic bone and the zygomatic process of the frontal bone. The

frontozygomatic suture between them may be palpated as a depression. Approximately 1 cm below this suture, a tubercle (Whitnall's) may be palpated within the orbital opening; it gives attachment to the lateral palpebral and check ligaments. The inferior border of the orbit is formed by the zygomatic bone laterally and the maxilla medially. It blends into the medial margin, which is less obvious, and is formed above by the frontal bone and below by the lacrimal crest of the frontal process of the maxilla. A shallow fossa behind the lower part of the medial wall houses the lacrimal sac. The anterior lacrimal crest is a guide to the location of the anterior ethmoidal artery, which may occasionally require ligation to control severe epistaxis, and which lies approximately 2.4 cm posterior to the crest. The optic canal lies approximately 4.5 cm posterior to the infraorbital margin (approximately 4.2 cm from the anterior lacrimal crest); these distances are important in orbital floor exploration in order to avoid optic nerve injury. The inferior orbital fissure lies approximately 2 cm posterolateral to the infraorbital margin; it does not contain any important structures and therefore surgery can be performed safely in this region. The infraorbital foramen, which transmits the infraorbital nerve and vessels, lies below the middle of the infraorbital margin. It commonly aligns with a vertical axis passing through maxillary tooth 5 ([Raschke et al 2013](#)) and sits an average of 2.5 cm (range 1.8–3.3 cm) from the midline face and 6.4 mm (range 3.2–12.1 mm) below the inferior orbital margin, with differences noted between genders and side ([Chrcanovic et al 2011](#), [Raschke et al 2013](#)). In neonates, the infraorbital nerve is located approximately half-way along a line drawn from the angle of the mouth to the midpoint of the palpebral fissure ([Bosenberg and Kimble 1995](#)).

The anterior surface of the maxilla is extensive and may be palpated between the infraorbital margin and the alveolar processes that bear the upper teeth. The canine eminence overlies the roots of the canine tooth and separates the incisive fossa anteriorly from the deeper canine fossa posteriorly. The bone of the canine fossa is thin and may be removed easily to access the maxillary sinus via an intraoral approach as part of a Caldwell–Luc procedure. The palatine process of the maxilla and palatine bone form the palpable roof of the mouth. The incisive/nasopalatine foramen sits in the midline palate, immediately posterior to the incisive papilla approximately 7.4 mm from the labial surface of the alveolar ridge ([Mraiwa et al 2004](#)) and transmits the sphenopalatine artery and nasopalatine nerve. The greater palatine foramen, which

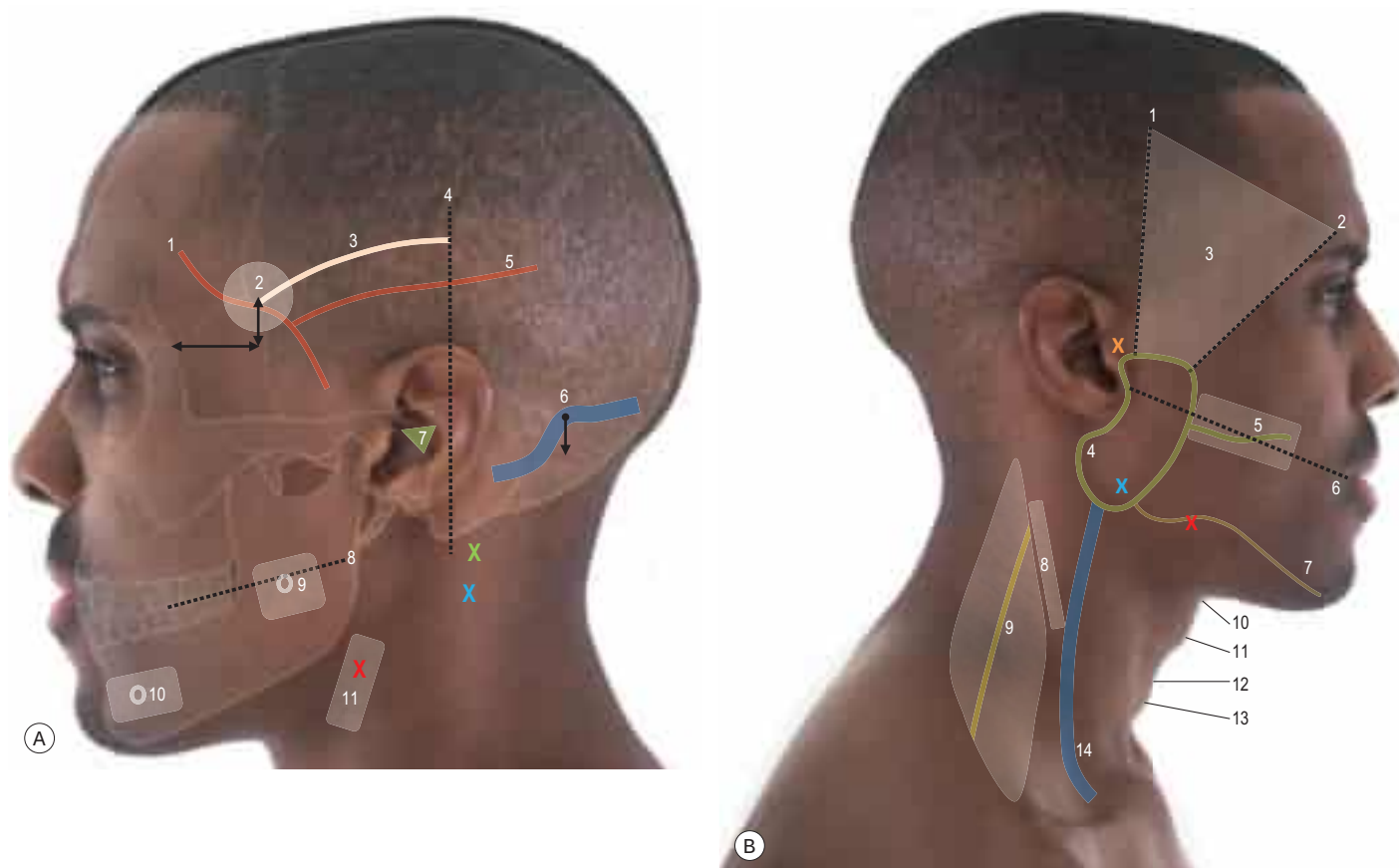


Fig. 26.8 The lateral aspect of the head and neck. **A**, Pterion, meningeal arteries, carotid bifurcation, mental and mandibular foramen. Key: 1, anterior branch of the middle meningeal artery: sits deep to the pterion in over half of subjects; 2, zone of location of the pterion: in a 1 cm diameter circle centred 2.5 cm posterior and 1.5 cm superior (black arrows) to the frontozygomatic suture; 3, lateral (Sylvian) fissure: aligns anteriorly with the anterior part of the squamosal suture in a zone approximately 3 cm anterior to the external acoustic meatus; 4, mastoid line; 5, posterior branch of the middle meningeal artery: runs parallel to the zygomatic arch and sits level with the supraorbital margin on the mastoid line; 6, asterion: sits almost always over or sometimes just below (black arrow) the transverse–sigmoid sinus junction; 7, suprameatal triangle; 8, occlusal plane; 9, zone of location (white) of the mandibular foramen: in the majority of cases, the foramen sits level with or below the occlusal plane (8), an average of 2 cm posterior to tooth 8; 10, zone of location (white) of the mental foramen: the foramen commonly sits between the roots of teeth 4 and 5 (range 3–6), usually level with or inferior to the tooth root apex; X (green), transverse process of C1 (atlas); X (blue), transverse process of C2 (axis); 11, zone of location (white) of the bifurcation of the common carotid artery: usually above the level of the thyroid lamina; X (red), most common level of common carotid bifurcation (C3). **B**, Parotid gland and duct, CN VII & XI, cutaneous branches of cervical plexus and internal jugular vein. Key: 1, porion line; 2, zygomatic sutural line; 3, zone of location of the (fronto)temporal nerve; 4, parotid gland; 5, zone of location (white) of the parotid duct (green line): sits within 1.5 cm of the middle half of a line passing from the lower tragus to the chelion; 6, lower tragus–chelion line; 7, marginal mandibular nerve; 8, zone of emergence of the cutaneous branches of the cervical plexus (white), posterior to the middle third of sternocleidomastoid; 9, zone of location of the accessory nerve, from 3–10 cm below the tip of the mastoid process to 1–10 cm above the insertion of trapezius into the clavicle; 10, hyoid bone; 11, thyroid cartilage (laryngeal prominence); 12, cricothyroid ligament/membrane; 13, cricoid cartilage; 14, external jugular vein; X (blue), angle of the mandible; X (red), facial artery and anterior border of masseter crossing the lower border of the mandible; X (orange), superficial temporal artery. (Adapted from Drake RL, Vogl AW, Mitchell A, et al (eds), *Gray's Atlas of Anatomy*, Elsevier, Churchill Livingstone. Copyright 2008.)

transmits the greater palatine artery and nerve, is commonly located in the region medial to maxillary tooth 8 (range: from level with tooth 7 to distal to tooth 8 (Sujatha et al 2005, Jaffar and Hamadah 2003) approximately 16 mm from the midline (Wang et al 1988).

Soft tissues and viscera

The muscular, fatty and cutaneous features that so clearly differentiate individuals are readily apparent on inspection. The external features of the eyelids and eyebrows are described on page 680 and page 475, respectively. The tympanic membrane may be examined under direct vision using an auroscope and is described on page 633. The retinal vascular supply may be examined directly by ophthalmoscopy.

Temporalis is palpable if the flat of the hand is placed over the temporal region and the jaw is clenched and relaxed. The anterior border of masseter is palpable and prominent when the jaw is clenched. The masseter is thicker in boys than in girls in both the relaxed and contracted state (Charalampidou et al 2008). With the mouth open, it is possible to examine the teeth, gingiva, tongue and the palatine tonsils lying between the faucial (palatine) arches formed by palatoglossus (anterolaterally) and palatopharyngeus (posteromedially). Peritonsillar abscess (quinsy) can cause ipsilateral tonsillar enlargement with contralateral displacement of the uvula, and parapharyngeal tumours also displace the ipsilateral tonsil and palate medially. The openings of the submandibular gland ducts can be seen on either side of the lingual

frenulum at its point of intersection with the sublingual mucosal folds. The lingual nerve often passes close to the lingual side of the root of mandibular tooth 8 (Behnia et al 2000), and can sit posteriorly on the retromolar ridge (Pogrel and Goldman 2004).

Parotid gland

The parotid gland is soft and indistinct to palpation. The superior border sits inferior to the zygomatic arch and passes from the mandibular condyle towards masseter. The anterior border passes inferiorly along the posterior border of masseter, often overlying its lateral surface, then over the mandibular angle to meet the anterior border of sternocleidomastoid, approximately 2 cm posteroinferior to the angle. The posterior border corresponds to a curve traced from the mandibular condyle to the mastoid process. The deep part of the gland lies within the prestyloid part of the parapharyngeal space. Tumours arising in this region present with medial displacement of the soft palate and tonsil. The parotid duct arises from the anterior margin of the gland and can be palpated on the anterior border of a clenched masseter. It passes along, or within 1.5 cm of, the middle half of a line drawn from the lower border of the tragus to the chelion (a bilateral landmark located at the outermost corner of the mouth where the upper and lower lips meet) (Stringer et al 2012) (see Fig 26.8B). The duct runs over masseter to its anterior border, and then bends sharply and pierces the underlying buccinator and buccal mucosa opposite maxillary tooth 7, where the duct orifice is visible and palpable. Duct cannulation is required for



Fig. 26.9 The anterior aspect of the face. Key: 1, zone of supraorbital foramen and neurovascular bundle (white) approximately 3 cm from the midline of the face; 2, zone of location of the infraorbital foramen and neurovascular bundle (white), approximately 2.5 cm from the midline of the face: aligns with a vertical axis passing through maxillary tooth 5; 3, vertical axis through maxillary tooth 5; 4, zone of location of the mental foramen and neurovascular bundle (white), approximately 2.5 cm (range, apices of teeth 3–6) from the midline of the face. (Adapted from Drake RL, Vogl AW, Mitchell A, et al (eds), *Gray's Atlas of Anatomy*, Elsevier, Churchill Livingstone. Copyright 2008.)

sialography, ductal dilation and basket retrieval of ductal stones (sialoliths).

Pulses

The facial artery can be palpated as it crosses the lower mandibular border immediately anterior to masseter and on the face approximately 1 cm lateral to the angle of the mouth. The superficial temporal artery is palpable immediately anterior to the tragus of the pinna. The artery divides into frontal and parietal branches approximately 2.5 cm above the zygomatic arch.

Facial nerve

The facial nerve may be injured in its extratemporal course as a result of facial trauma and laceration, or iatrogenically during surgery involving the parotid or submandibular glands or the temporomandibular joint, or during facelift procedures. Knowledge of the topographical and surgical landmarks is therefore essential if the nerve is to be preserved during surgery.

The nerve exits the skull at the stylomastoid foramen and so is initially deep to the posterior margin of the external acoustic meatus; the nerve trunk lies approximately 1 cm inferomedial to the tip of the tragal pointer, and runs anterior to the mastoid apex and C1 transverse process, within 9 mm and 14 mm, respectively (Greyling et al 2007). It usually divides within the parotid gland into five main branches that radiate across the face (Fig. 26.10; see Fig. 30.24). Cadaveric and intra-operative microdissection studies have revealed that the course of its branches is highly variable and therefore surface coordinates only approximate their trajectory. The temporal/frontotemporal branches emerge from the superior surface of the parotid, cross the zygomatic arch and travel towards the pterion and the superolateral brow. The majority pass in the zone between two lines: line 1 (porion line) joins two points each situated 12 mm from the porion on both the supra-orbitomeatal line and the infraorbitomeatal line; and line 2 (zygomatic sutural line) is a line joining the zygomaticotemporal and frontozygomatic sutures (Davies et al 2012). These branches may be injured in facial rhytidectomy, coronal or endoscopic brow lifting, and temporal craniotomy. The buccal branches run both above and below the parotid duct. The marginal mandibular nerve emerges from the anteroinferior parotid and passes anteriorly. When posterior to the facial artery, the nerve almost always lies inferior to the lower border of the mandible by an average of 1.2 cm (range 0.9–1.75 cm) (Batra et al 2010). The branches that enter the neck re-enter the face by crossing the lower

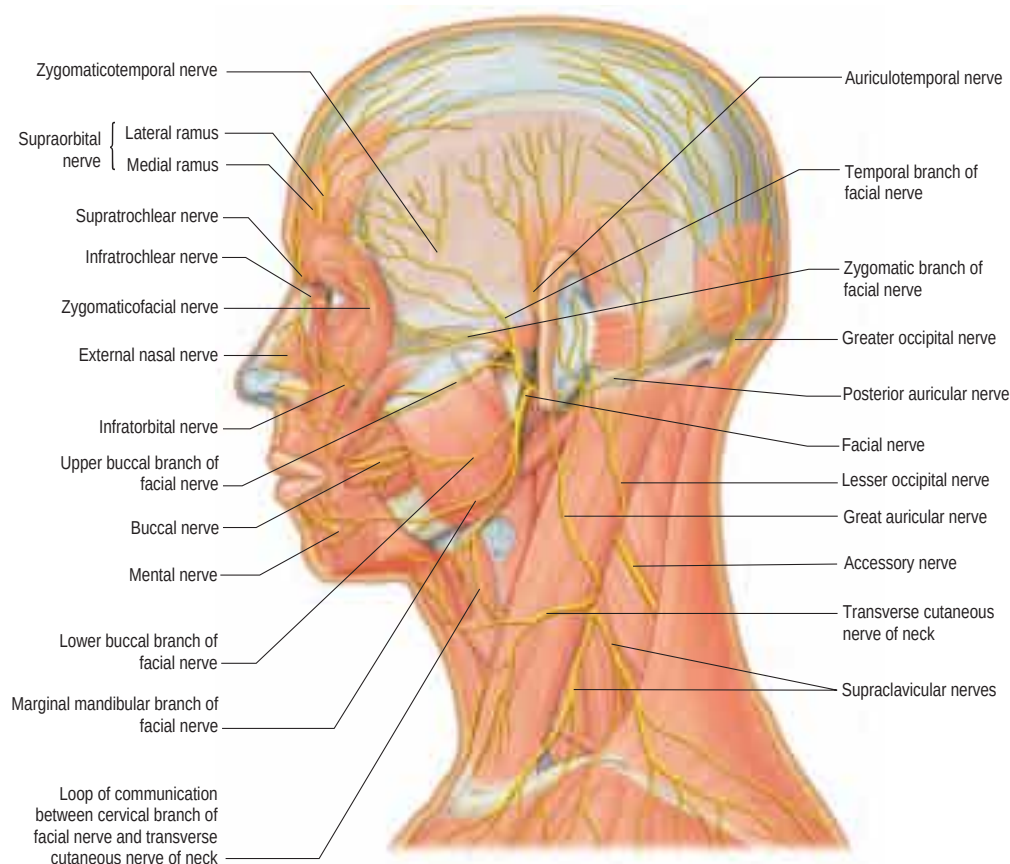


Fig. 26.10 The distribution of the cutaneous branches of the trigeminal nerve and of the cervical plexus, and the motor branches of the facial nerve in the face and neck.

border of the mandible at the anterior border of masseter and pass superficial to the facial artery and vein. Regional incisions are therefore made at least 1.5 cm below the lower mandibular border to avoid nerve injury, which results in an upturned ipsilateral mouth via paralysis of depressor anguli oris and depressor labii inferioris (Chowdhry et al 2010).

Dermatomes

Our knowledge of individual dermatome position is based on clinical evidence. The dermatomes of the face arise mainly from cutaneous branches of the three major divisions of the trigeminal nerve (see Fig. 26.10). The skin covering the front and sides of the neck, and over the angle of the mandible, and over the lateral scalp and posterior aspect of the pinna is supplied by branches of the cervical plexus (C2–C4), and the skin over the back of the head is supplied by the greater occipital nerve (C2, posterior primary ramus) (see Figs 26.10, 30.22, 43.3 and 43.76).

SURFACE ANATOMY OF THE NECK

Skeletal and cartilaginous landmarks

The neck extends from the pericraniocervical line superiorly to the level of the clavicle, scapula and thoracic inlet (first rib and superior manubrium) inferiorly, where it is continuous with the thoracic cavity and upper limb. The spines of the second and seventh cervical vertebrae are the most prominent and may be palpated in the midline of the posterior neck (the former via deep palpation). The remaining cervical spines are indistinct because they are covered by the ligamentum nuchae, on either side of which lie the masses of the postvertebral musculature. The transverse process of the first cervical vertebra is palpable in the hollow region posteroinferior to the mastoid apex, and deep palpation inferior to this reveals the transverse process of the second cervical vertebra.

The hyoid bone body can be felt in the midline neck a few centimetres below and behind the chin, at the junction of the skin of the anterior neck and the floor of the mouth. The greater horns may be palpated between a finger and thumb, and moved from side to side. In the adult, the body of the hyoid bone sits level with the fourth cervical vertebra (range C3–C5/6) (Mirjalili et al 2012b); it is higher in children. The laryngeal prominence (Adam's apple) of the thyroid cartilage is palpable in the midline neck, and the thyroid laminae can be felt passing posterolaterally. In the male, the prominence is usually visible, whereas in the female it is not usually apparent. The curved upper border of the thyroid cartilage, the midline thyroid notch and the superiorly located depression of the thyrohyoid membrane/ligament are easily palpable. The upper border of the thyroid cartilage usually lies between the fourth and fifth cervical vertebrae (Mirjalili et al 2012b). The firm, smooth anterior arch of the cricoid cartilage is palpable below the inferior border of the thyroid cartilage. The inferior border of the cricoid commonly sits at the level of the seventh cervical vertebra (range C5–T1); the posterior cricoid overlaps the sixth cervical vertebra (Mirjalili et al 2012b). The indentation between the thyroid and cricoid cartilages is covered by the anterior/median cricothyroid ligament, a useful site for emergency access to the airway in cases of obstruction at or above the vocal cords (cricothyroid puncture or cricothyroidotomy). In the neonate, the dimensions of the cricothyroid membrane are too small to allow safe passage of an airway (Navsa et al 2005). Confirmation of the surface anatomy of the hyoid bone, thyroid and cricoid cartilages, and suprasternal notch is essential prior to performing a tracheostomy (Ellis et al 2004).

The clavicle is a sigmoid-shaped bone that is easily visible in thin people and is palpable in all except the morbidly obese. Its medial two-thirds are rounded and convex anteriorly and the lateral third is flat and concave anteriorly. The suprasternal (jugular) notch lies between the expanded medial ends of the clavicles; the superior edge of the sternal manubrium forms its inferior border. For much of its length, the clavicle may be almost encircled by two fingers, but medially its ligamentous attachments make definition more difficult. The posterior end of the first rib may be felt indistinctly within the floor of the posterior triangle of the neck.

Soft tissues and viscera

Triangles and root of the neck

The neck is divided into anterior and posterior triangles by sternocleidomastoid (Fig. 26.11). In cadaveric specimens, the anterior triangle may

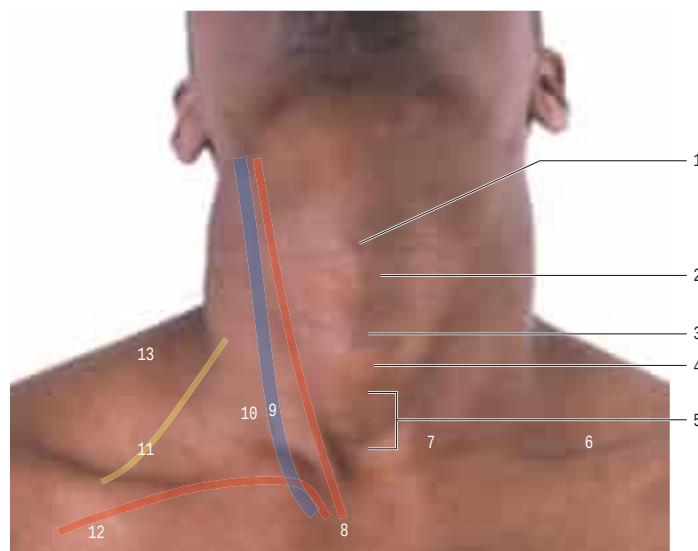


Fig. 26.11 The anterior and posterior triangles of the neck. Key: 1, thyroid notch with thyrohyoid ligament (indented region) above; 2, laryngeal prominence; 3, median cricothyroid ligament; 4, cricoid cartilage; 5, isthmus of thyroid gland; 6, greater supraclavicular fossa (of posterior triangle); 7, sternal head of sternocleidomastoid; 8, common carotid artery; 9, lesser supraclavicular fossa overlying the internal jugular vein; 10, clavicular head of sternocleidomastoid; 11, course of brachial plexus (upper trunk); 12, subclavian artery; 13, trapezius. (Adapted from Drake RL, Vogl AW, Mitchell A, et al (eds), Gray's Atlas of Anatomy, Elsevier, Churchill Livingstone. Copyright 2008.)

be further subdivided into submental, muscular, carotid and digastric triangles, and the posterior triangle may be divided into occipital and supraclavicular triangles. Above the hyoid bone, the musculature runs in a predominantly horizontal or oblique direction, and below the hyoid it runs in a vertical direction. The muscles forming the boundaries of the subtriangles of the neck are not readily palpable or visible, although they may be landmarked according to their attachment points, e.g. the two bellies of digastric lie along a line from the anterior midline chin to the lateral aspect of the hyoid body and then towards the medial part of the mastoid process.

The root of the neck is the transitional zone between the neck, thorax and both upper limbs. Communication with the thorax via the thoracic inlet is marked anteriorly by the superior manubrium. The trachea is palpable in the midline superior to the sternal notch, a region that may also contain the left brachiocephalic vein. The apices of the lungs lie posterior to the medial third of the clavicle (Mirjalili et al 2012a).

Anterior triangle

The base of the anterior triangle is formed by the inferior border of the mandible and a line from its angle to the mastoid process. The sides are formed by the midline anteriorly and by the anterior border of sternocleidomastoid laterally. The triangle is best palpated bimanually with the examiner standing behind the subject and using both hands to examine the structures within. Inspection reveals the rounded tendinous sternal head of sternocleidomastoid, which arises from the superolateral angle of the manubrium, and the broader, more vertical clavicular head, which arises from the upper surface of the medial third of the clavicle. The indentation of the lesser supraclavicular fossa sits between the two heads, anterior to the internal jugular vein before it joins the subclavian vein posterior to the sternoclavicular joint; the fossa therefore represents an access point for central venous cannulation.

The trachea and its cartilaginous 'rings' may be palpated in the midline inferior to the cricoid cartilage and above the sternal notch. It may be deviated laterally by thyroid masses/growth, atelectasis or tension pneumothorax. The thyroid gland consists of two soft, palpable lobes lying on either side of the lower half of the laminae of the thyroid cartilage and upper part of the trachea, their lower borders extending towards the sternal end of the clavicle on each side. Swallowing may assist palpation of the gland. The lobes are joined by the isthmus (approximately 2 cm wide), which sits anterior to the second and third tracheal cartilages; these cartilages may therefore be impalpable.

Apart from a few retrovisceral nodes and some nodes deep to sternocleidomastoid, all head and neck lymph nodes groups are clinically

palpable when enlarged. Lymph nodes above the hyoid bone tend to be disposed in a horizontal plane either on or below the pericranio-cervical line, and in front of and behind the pinna, whereas the deep cervical nodes run vertically and are related to the internal jugular vein. The jugulodigastric node group are palpable in the indented region posterior to the mandibular angle and anterior to sternocleidomastoid. Palpation medial and deep to the inferior border of the mandible can reveal nodes within the submental and submandibular triangles. Superficial cervical nodes lie adjacent to the external and anterior jugular veins and are landmarked with them.

The submandibular gland is palpable within the submandibular triangle. It extends approximately 2 cm beneath the lower border of the posterior part of the body of the mandible and may reach the greater horn of the hyoid bone. Its posterior border is level with the angle of the mandible and its anterior border extends approximately 4 cm anteriorly.

Posterior triangle

The posterior triangle is bounded by the posterior border of sternocleidomastoid, the middle third of the superior surface of the clavicle (the base), and the anterior margin of trapezius. Its apex is the point where sternocleidomastoid and trapezius approximate at the superior nuchal line. The lower portion of the posterior triangle forms the greater supraclavicular fossa, an important clinical area that lies just above and behind the clavicle at the confluence of the thoracic inlet and the aditus to the axilla and arm. It is best inspected from in front but palpated from behind. Scalenus anterior and medius are palpable as firm masses in the floor of the triangle, posterior to the lower part of sternocleidomastoid. The near-vertical interscalene groove is palpable between them; at or near the level of the sixth cervical vertebra, it serves as a landmark for the upper trunk of the brachial plexus.

When the greater supraclavicular fossa is inspected, the pulsation of the great veins may be seen if the central venous pressure is raised. The fossa is also a common site in which to feel pathologically enlarged supraclavicular lymph nodes. In particular, cancers of the lung can spread to left- and right-sided nodes, and cancers of the upper gastrointestinal tract can spread to left-sided nodes. The subclavian artery may be felt pulsating as it crosses the first rib, posterior to the medial end of the clavicle, and the upper and middle trunks of the brachial plexus may be felt above and behind it. The posterior end of the first rib may be felt as a fullness in the posterior aspect of the fossa. A point approximately 2.5 cm above the middle of the medial third of the clavicle marks the level of the neck of the first rib and is thus the surface marking for the apex of the dome of the cervical pleura and lung covered by the suprapleural membrane (Sibson's fascia). Any penetrating injury (iatrogenic or otherwise) in this region will run the risk of creating a pneumothorax.

Course of vessels

Carotid arteries

In the neck, the common carotid artery and its continuation, the internal carotid artery, may be represented by a more or less straight line passing from the sternoclavicular joint to a point just behind the mandibular condyle. The level at which the common carotid artery bifurcates into internal and external carotid arteries is variable (Lo et al 2006). It ranges from the first to the fifth cervical vertebra (Furukawa et al 2012), and is found most commonly at the level of the third cervical vertebra (Furukawa et al 2012, Mirjalili et al 2012b). Bifurcations almost always sit above the upper border of the thyroid cartilage lamina by an average of 1.6 cm (Mirjalili et al 2012b). Almost half occur at the superior border of the thyroid cartilage, and less frequently opposite the hyoid bone or between the thyroid cartilage and hyoid bone (Al-Rafiah et al 2011). The common carotid artery may be compressed against the prominent transverse process of the sixth cervical vertebra (Chassaignac's or carotid tubercle), which sits lateral to the cricoid cartilage. Above this level, the artery is superficial and its pulsation may be readily felt beneath the anterior border of sternocleidomastoid.

Subclavian artery

The subclavian artery enters the root of the neck behind the sternoclavicular joint. It passes through the inferior part of the palpable interscalene groove and arches upwards to reach a point approximately 2 cm above the clavicle deep to the posterior border of sternocleidomastoid, before passing across the upper surface of the first rib behind the middle of the clavicle. The subclavian pulse may be detected behind the clavicle at the lateral border of sternocleidomastoid or where it crosses the first rib.

Subclavian vein

The subclavian vein sits anterior to the subclavian artery and follows a similar course by passing over the first rib into the root of the neck. It joins the internal jugular vein to form the brachiocephalic vein posterior to the sternoclavicular joint (Mirjalili et al 2012a) and lies closest to the clavicle at the junction of its middle and medial thirds, approximately 7 cm from the midline (Hale et al 2010). Subclavian vein cannulation is described in Chapter 29.

Anterior jugular vein

The anterior jugular vein runs downwards from beneath the chin, approximately a finger's breadth from the midline. It turns laterally, approximately 2.5 cm from the sternal end of the clavicle, and passes beneath sternocleidomastoid to drain into the external jugular vein. The veins of each side join to form a jugular arch just above the manubrium sterni. The anterior jugular vein may be encountered if the operator drifts laterally while creating a surgical airway.

External jugular vein

The external jugular vein lies superficial to sternocleidomastoid and may be represented by a line that starts just below and behind the angle of the mandible and runs to a point close to the lateral extent of the junction of sternocleidomastoid with the clavicle. It drains into the subclavian vein after penetrating the investing layer of deep cervical fascia and may be kinked at this point. If the proximal part of the vein is damaged, it may be held open by the surrounding fascia; air may then be sucked in, resulting in an air embolus. The external jugular vein can be distended if venous pressure is raised, e.g. by performing a Valsalva manoeuvre (forced expiration against a closed glottis and mouth, and blocked nostrils), by supraclavicular digital pressure or if the vessel is kinked.

Internal jugular vein

The internal jugular vein runs in the carotid sheath, lying just lateral or anterolateral to the pulsating carotid arteries. It therefore has similar surface markings to those described for the common and internal carotid arteries, and is represented by a broad band from the lobule of the ear to the sternoclavicular joint, where it joins the subclavian vein (Mirjalili et al 2012a). The inferior bulb of the internal jugular vein lies in the lesser supraclavicular fossa (the depression between the sternal and clavicular heads of sternocleidomastoid), where it may be accessed for central vein cannulation. In children under the age of 5 years, the vein is less than 1 cm deep to skin (Roth et al 2008). The jugular venous pulse reflects the jugular venous pressure (central venous pressure) and is a measure of cardiac function. Pulsation of the great veins may be seen in this region if the central venous pressure is raised (e.g. right-sided heart failure). Internal jugular vein cannulation is described in Chapter 29.

Course of nerves

Accessory nerve

The accessory nerve may follow a straight or tortuous course; descriptions vary (Symes and Ellis 2005, Mirjalili et al 2012c). Its course has traditionally been approximated by a line passing across the floor of the posterior triangle, from the junction of the upper and middle thirds of the posterior border of sternocleidomastoid (Erb's point; see Fig. 26.11) to the junction of the lower and middle thirds of the anterior border of trapezius. However, morphometric studies have demonstrated considerable positional variability. The nerve emerges from the posterior border of sternocleidomastoid approximately 6.5 cm below the tip of the mastoid process, and penetrates the anterior border of trapezius approximately 4.5 cm above the clavicular insertion of trapezius (Symes and Ellis 2005; Mirjalili et al 2012c) (Figs 26.8B, 26.10). Accessory nerves usually divide into 2–4 branches prior to trapezius (Mirjalili et al 2012c).

Cervical plexus

The cutaneous branches of the cervical plexus emerge from behind the posterior border of the middle third of sternocleidomastoid, usually below the accessory nerve (see Figs 26.8B, 26.10). The great auricular nerve can be damaged during rhytidectomy. It passes superiorly over sternocleidomastoid towards the lobule and angle of the mandible; it can sit over the mid-belly of sternocleidomastoid, approximately 6.5 cm below the lower border of the external acoustic meatus, with the external jugular vein sitting 0.5 cm anteriorly at this position (McKinney and Katrana 1980).

Brachial plexus

The roots and trunks of the brachial plexus can be represented by a line passing between the middle of the posterior border of sternocleidomastoid and the middle of the clavicle. In thin individuals, the scalene muscles are visible when the head is laterally flexed to the contralateral side; the roots/upper trunk are palpable in the almost vertical inter-scalene groove, which is located posterior to scalenus anterior. The trunks lie in the posterior triangle of the neck, and the divisions lie behind the clavicle near the lateral border of the first rib. The cords enter the axilla above the first part of the subclavian artery, embrace its second part and give off branches around the third part.

Cervical sympathetic ganglia

The superior cervical ganglion lies slightly anterior to the (palpable) transverse process of the second cervical vertebra, while the middle cervical ganglion, when present, lies just in front of the transverse process of the sixth cervical vertebra, the tubercle of which is palpable via the anterior neck, lateral to the cricoid cartilage. The inferior cervical ganglion may be fused with the first thoracic cervical ganglion, forming the stellate or cervicothoracic ganglion. Stellate ganglion block is often employed to perform a sympathetic nerve block to the head and neck, or to the arm (Ellis et al 2004).

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The intimate relationship of the lingual nerve to the lower third molar region is a significant factor in the morbidity associated with third molar extractions. This extensive cadaver study reported that in 14% of patients the lingual nerve was located above the lingual crest and that in 22% the nerve was in direct contact with the lingual plate of the alveolar process.

Day JD, Tschabitscher M 1998 Anatomic position of the asterion. *Neurosurgery* 42:198–9.

The asterion has been used as a landmark in lateral approaches to the posterior fossa. It is not strictly reliable in terms of locating the underlying posterior fossa dura but it does lie directly over the transverse–sigmoid sinus complex. Caution must therefore be taken when placing burr-holes to avoid damage to the venous sinuses with potentially very serious consequences.

Hale SJ, Mirjalili SA, Stringer MD 2010 Inconsistencies in surface anatomy: the need for an evidence-based reappraisal. *Clin Anat* 23:922–30.

An argument that an evidence-based framework is essential if surface anatomy is to be accurate and clinically relevant.

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